Using the Municipal-to-Treasury Ratio to Identify Relative Value Opportunities

Aikyan Vadlamani
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ABSTRACT

This study examines whether the Municipal-to-Treasury ratio can be used to identify when AAA municipal bonds offer better relative value compared with U.S. Treasury bonds. Using daily 10-year Triple-A Municipal Market Data (MMD) and U.S. Treasury yield data, I analyzed how the ratio changed across different market environments from 2006 through 2025. I first tested a 180-day rolling average and a long-term historical average, but both were affected by major market disruptions and did not always provide a useful benchmark. I therefore divided the data into four five-year periods and compared the ratio with the average for each period. The results showed that the Muni-to-Treasury ratio ("muni-ratio") increased sharply during major market events, especially the Great Financial Crisis (GFC) and the COVID-19 pandemic, and generally moved back toward its average over time. However, the amount of time needed for mean reversion varied across the four periods. The results also showed that a high muni-ratio did not always lead to higher investment returns because municipal and Treasury yields could move together while the ratio stayed relatively unchanged. Overall, the findings support the idea that the muni-ratio can be a useful tool for identifying relative value opportunities, but it should not be used by itself to predict future returns or the exact timing of an investment.

1. Introduction

1.1 Background

When examining investment options, there exists a variety of asset categories to select from. The main categories include equities and fixed income like bonds. This research paper will center on bonds, particularly municipal bonds, and will explore how municipal bonds compare to Treasury bonds from a relative-value perspective, using the municipal-to-Treasury ratio (muni-ratio) as a measure. Professional investors use the muni-ratio as an indicator of relative value between municipal bonds and Treasury bonds. If the muni-ratio exceeds its historical average, municipal bonds may present more favorable value compared to Treasury bonds. Traditionally, older retired investors and individuals with substantial wealth have favored the municipal market due to the tax-exempt nature of interest earned on municipal bonds, which generally exempts them from federal income taxes on those gains.

The Treasury bond market and municipal bond market had estimated sizes of around $31 trillion and $4.5 trillion, respectively, at the end of Q1 2026.[1,2]

A municipal bond, or muni, represents a loan to a state government, local government, or nonprofit entity and is used to finance essential infrastructure such as roads, hospitals, water systems, nonprofit universities, and public schools. High-quality municipal bonds, like Treasury bonds, are generally viewed as safer fixed-income investment options relative to corporate bonds. Notably, interest income from Treasury bonds is taxable, unlike muni bond interest income, which is tax-exempt. In my analysis, I considered a 30% tax rate. In the upcoming sections, I will elaborate on the reasoning behind selecting that rate.

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[1] Securities Industry and Financial Markets Association. (2026, August 10). U.S. Treasury securities statistics. SIFMA. https://www.sifma.org/research/statistics/us-treasury-securities-statistics

[2] Securities Industry and Financial Markets Association. US Municipal Bond Statistics

1.2 Research Question

The main research question is: Can the Municipal-to-Treasury Ratio ("muni-ratio") identify periods when the AAA municipal bonds offer better relative value and subsequently outperform comparable Treasury bonds? My hypothesis is that when the muni-ratio is significantly higher than its five-year average, municipal bonds may offer better relative value and may subsequently outperform comparable Treasury bonds.

Table 1 shows the inverse relationship between the yield and price of a 10-year bond, with a face value of $100 and a 5% coupon.

Price	Coupon	Yield
$100	5%	5%
$90	5%	6.37%
$110	5%	3.79%

1.3 Significance of the Analysis

The research I reviewed prior to my analysis, focused mainly on factors that affect municipal bond prices and yields, with less attention given to whether the muni-ratio can identify future relative value opportunities. I believe that it is an interesting question to investigate using mathematical models and historical market data. Bond prices and investment returns can be influenced by many factors, including interest rates, economic conditions, credit quality, tax policy, government spending, and investor demand. Municipal bonds are affected by these

factors as well. For example, changes in federal tax rates can alter the tax advantage of municipal bonds, while changes in government spending and economic policy can affect interest rates and the financial condition of state and local governments.

Major economic and political events can also create uncertainty in the market and cause investors to move toward or away from bonds. Because these factors can change the relationship between municipal and Treasury yields, looking at different five-year periods can help show whether the muni-ratio remains a useful indicator of relative value across different market environments.

2. Literature Review

The muni-ratio is a metric investors use to compare municipal bond yields to U.S. Treasury yields, and the muni-ratio is often treated as a predictive indicator of relative value between municipal bonds and Treasury bonds. The muni-ratio could move for a variety of reasons that include municipal bond supply and demand, market stress or dislocations, muni market-related headlines and their impact on retail demand for muni bonds, local economic conditions, and tax policy.

Even though not many studies examine the muni-ratio as an indicator of relative value, there is plenty of practitioner research showing that muni-to-Treasury relationships can contain useful clues about future returns, which is why the ratio is worth analyzing in an academic research paper.

Cestau et al. (2019)[3] describe the municipal bond market as the largest and most important market system in the US, with over $4 trillion in market capitalization, 55,000 issuers,

[3] Cestau, D., Hollifield, B., Li, D., & Schürhoff, N. (2019). Municipal bond markets. Annual Review of Financial Economics, 11, 65-84. https://doi.org/10.1146/annurev-financial-110118-123034

1.5 million bond issues, and 100,000-200,000 new bonds issued each year. This article explores how financial crises heavily affect municipal bonds and highlights the complex nature of the municipal bond market.

Babina et al. (2021)[4] describe how taxes affect municipal bond pricing and the economic forces behind it. This article helps improve the understanding of municipal bond ownership and pricing, specifically how investors look for tax benefits based on different preferences, such as yield, maturity, and marginal tax rate. The marginal tax rate is the tax rate applied to the very last dollar of income that is taxable.

A financial crisis is an event that can drastically affect the relationship between municipal and Treasury bond yields. Li and Lu (2020)[5] examined the impact of the COVID-19 pandemic on the municipal bond market. The article shows how the muni market helps local and state governments function effectively. Li and Lu (2020)[6] discuss the impact of COVID-19 and found that in the past decade, more than 50,000 municipal issuers sold an average of $435 billion in new debt per year. The study shows that, during the pandemic, the offering yields of municipal bonds increased with the severity of COVID-19 cases in a given state.

Li and Lu (2020)[7] note that the number of bond issuances decreased substantially, to levels seen in 2008 and 2009 during the Great Financial Crisis (GFC). Furthermore, the study finds that the actions of the Federal Reserve impact the ability of municipal bonds to adjust to changes in interest rates. These findings confirm that financial crises can affect the municipal bond market in unique ways. Furthermore, the authors discover that with the rise in COVID-19

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[4] Babina, T., Jotikasthira, C., Lundblad, C., & Ramadorai, T. (2021). Heterogeneous Taxes and Limited Risk Sharing: Evidence from Municipal Bonds. The Review of Financial Studies, 34(1), 509-568. https://doi.org/10.1093/rfs/hhaa028

[5] Li, Tao, and Lu, Jing. *Municipal Finance During the COVID-19 Pandemic: Evidence from Government and Federal Reserve Interventions* (June 28, 2020). https://ssrn.com/abstract=3637636

[6] Li, Tao, and Lu, Jing. Municipal Finance During the COVID-19 Pandemic

[7] Ibid

cases and fatalities, the yield of munis tends to increase, whereas their issuance decreases. Moreover, they find that local emergency declarations are correlated with a rise in the yield of munis by approximately 69 basis points. The authors conclude that investors during the pandemic were careful regarding the bonds they purchased and avoided transportation bonds, bonds tied to specific taxes, and bonds from weaker states.

3. Methodology

3.1 Data

This study used daily yield data for the 10-year Triple-A Municipal Market Data (MMD)[8] curve and the 10-year U.S. Treasury yield[9,10,11,12]. The MMD curve represents the benchmark yield for high-quality AAA municipal bonds, while the Treasury yield represents the yield on U.S. government securities with the same maturity. Using the same 10-year maturity made it possible to compare the two markets more consistently.

The main measure used in the analysis was the muni-ratio. The ratio was calculated by dividing the 10-year MMD yield by the 10-year Treasury yield:

Muni-to-Treasury Ratio ("muni-ratio") = 10-Year MMD Yield ÷ 10-Year Treasury Yield

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[8] LSEG, "The Municipal Market Monitor (TM3)," LSEG, https://www.lseg.com/en/data-analytics/trading-solutions/municipal-market-monitor-tm3

[9] U.S. Department of the Treasury, 1-2-2024, "Daily Treasury Rates," https://home.treasury.gov/resource-center/data-chart-center/interest-rates/TextView?type=daily_treasury_yield_curve&field_tdr_date_value=2024

[10] U.S. Department of the Treasury, 1-2-2025, "Daily Treasury Rates," https://home.treasury.gov/resource-center/data-chart-center/interest-rates/TextView?type=daily_treasury_yield_curve&field_tdr_date_value=2025

[11] U.S. Department of the Treasury, 1-2-2026, "Daily Treasury Rates," https://home.treasury.gov/resource-center/data-chart-center/interest-rates/TextView?type=daily_treasury_yield_curve&field_tdr_date_value=2026

[12] U.S. Department of the Treasury, 12-29-2023, "Daily Treasury Rate Archives," https://home.treasury.gov/resource-center/data-chart-center/interest-rates/daily-treasury-rate-archives

The result was expressed as a percentage. For example, a ratio of 80% means that the AAA municipal bond yield is equal to 80% of the yield on a comparable Treasury bond.

Because municipal bond interest is generally exempt from federal income taxes while Treasury interest is federally taxable, I also considered the effect of taxes when comparing the two investments. For this study, I assumed a 30% federal income tax rate. This was used as a simplified example and does not represent the tax rate of every investor.

3.2 Selecting a Benchmark

One of the main goals of the study was to determine whether the muni-ratio was unusually high or low compared with its normal level. I first compared the ratio with a 180-day rolling average. However, I found that major market events could cause the rolling average itself to rise sharply and remain elevated even after the original market disruption had passed. This made the rolling average less useful as a stable benchmark.

I then tested a historical average using the full dataset. The long-term average was also not ideal because the normal level of the muni-ratio changed significantly over time. For example, ratios during earlier periods were generally much higher than those seen in more recent years.

For this reason, I divided the data into four five-year periods:

- 2006-2010
- 2011-2015
- 2016-2020
- 2021-2025

I calculated a separate average muni-ratio for each five-year period. This allowed the ratio to be compared with a benchmark that better reflected the interest-rate and market environment of that particular period.

3.3 Identifying Market Dislocations

After calculating the average ratio for each five-year period, I looked for periods when the muni-ratio moved significantly above its average. These increases were treated as possible market dislocations because municipal yields had become unusually high compared to Treasury yields.

The main dislocation windows examined were:

- November 24, 2008 to January 20, 2009
- June 1, 2011 to January 1, 2012
- March 9, 2020 to June 5, 2020
- January 1, 2022 to December 31, 2023

These periods were selected because they contained noticeable increases in the muni-ratio within each five-year period. They also occurred during different economic and market environments, allowing the behavior of the ratio to be compared across several types of market conditions.

3.4 Calculating Forward Returns and Excess Returns

To understand whether a high muni-ratio was followed by better municipal bond performance, I calculated forward returns after selected dates during each market dislocation period. For the municipal side, I estimated the return on a hypothetical AAA municipal bond priced using the MMD benchmark yield. I then compared those returns with the returns on a comparable Treasury bond. A hypothetical $1 million investment was used for each so that the results could be compared in dollar terms.

For the bond pricing calculation, I used a 10-year bond and compared the yield on the starting date with the yield at a later date. Because bond prices and yields move in opposite directions, a decrease in yield causes the price of an existing bond to increase, while an increase in yield causes its price to decrease. I used this relationship to estimate how much the value of the $1 million investment changed over each holding period.

For Figure 7, I looked at the returns after one month, two months, and three months from the selected starting dates. For each holding period, I calculated the municipal bond return and the Treasury bond return separately. This allowed me to see not only whether the municipal bond produced a positive return, but also whether it performed better than the Treasury bond over the same period.

I then calculated the excess return, which measures how much better or worse the municipal bond performed compared with the Treasury bond:

Excess Return = Municipal Bond Return - Treasury Bond Return

For example, if the municipal bond earned a 3% return and the Treasury bond earned a 1% return, the municipal bond had an excess return of 2 percentage points. For a $1 million investment, this would mean that the municipal investment earned approximately $20,000 more than the Treasury investment over the same period.

A positive excess return meant that the municipal bond outperformed the Treasury bond, while a negative excess return meant that the Treasury bond performed better. This calculation was especially important for my research question because the goal was not only to see whether municipal bonds earned a positive return, but whether they offered better relative performance than Treasury bonds after the muni-ratio moved above its five-year average.

Looking at excess returns also helped separate relative value from absolute returns. A municipal bond could have a negative return and still outperform a Treasury bond even if the Treasury bond lost more money. In the same way, both investments could have positive returns while the Treasury bond performed better. Therefore, the excess return provided a clearer way to test whether a high muni-ratio actually identified a relative value opportunity.

3.5 Measuring Mean Reversion

I also examined whether the muni-ratio moved back toward its five-year average after a major increase. This process is referred to as mean reversion. For each selected observation, I measured the amount of time between the starting date and the date when the muni-ratio returned to its five-year period average. I recorded this time in days.

This part of the analysis was important because identifying an attractive investment opportunity is different from knowing when that opportunity will produce a return. If the ratio takes a long time to return to its average, an investor may have to hold the investment for a longer period before the relative-value opportunity is realized.

The mean-reversion results were then compared across the four five-year periods to determine whether the speed of mean reversion was consistent or depended on the market environment.

3.6 Software and Analysis

Google Colab was used for the coding portion of this analysis. Python code was used to organize and clean the data, calculate the muni-ratio, create tables and graphs, measure changes in yields, and analyze the time it took for the ratio to return toward its average. ChatGPT and Claude were used as tools to help write and review parts of the code. The results were checked throughout the process by comparing the calculations with the original data and reviewing the outputs to make sure the code was producing the intended results.

4. Results

At the beginning of my analysis, I used a 180-day rolling average as the benchmark for the Muni-to-Treasury ratio, as shown in Figure 1. However, after major market events, the rolling average could remain elevated and sometimes peak several months after the original spike in the ratio. This made the rolling average less useful as a benchmark. I then compared the ratio with its long-term historical average, which was approximately 87%, as shown in Figure 2.

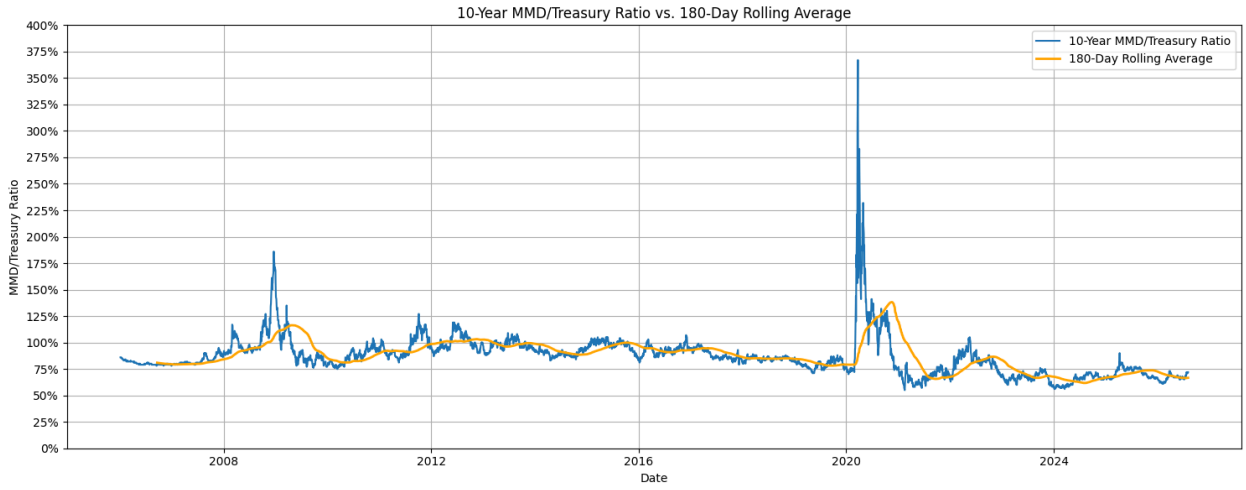


Figure 1: Municipal-to-Treasury Ratio Relative to Its 180-Day Rolling Average

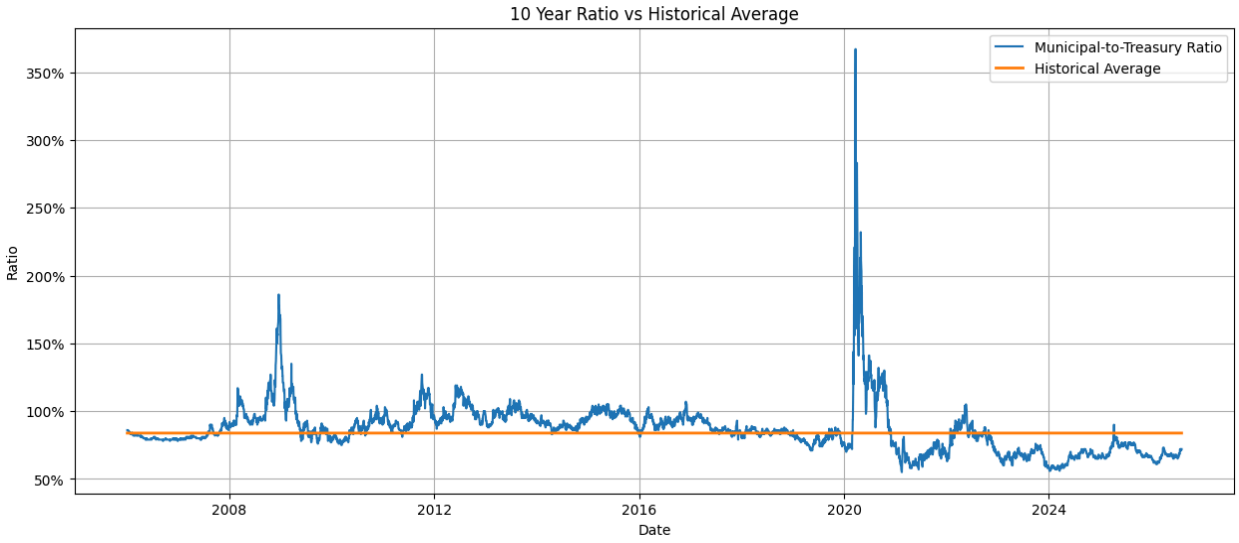


Figure 2: Municipal-to-Treasury Ratio Relative to Its 20-Year Historical Average

However, this average also did not represent every market period equally well. The average ratio from 2006 through 2020 was above 90%, while the average from 2021 through 2025 was closer to 70%. Because of this difference, I divided the data into four five-year periods: 2006-2010, 2011-2015, 2016-2020, and 2021-2025. A separate average was calculated for each period so that the ratio could be compared with a benchmark that better reflected the market environment at the time.

The four five-year periods and their averages are shown in Figures 3 through 6. The results showed that the Muni-to-Treasury ratio behaved differently during each period. The earlier periods generally had higher average ratios and larger spikes, while the 2021-2025 period had an average closer to 70%. As of August 25, 2026, the ratio was approximately 68%, which

was much closer to the average noted during the most recent five-year period than the averages seen in earlier periods.

Figures 3 through 6 below show several periods when the muni-ratio moved significantly above its five-year average. These periods were important because they represented possible relative value opportunities where municipal yields became unusually high compared to Treasury yields.

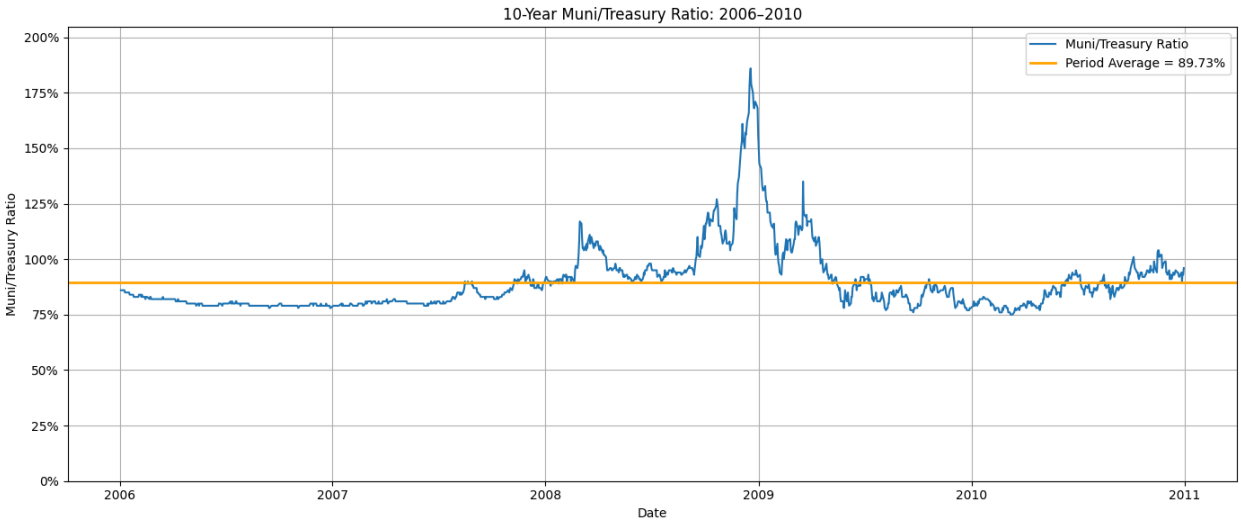


Figure 3: Municipal-to-Treasury Ratio, 2006-2010, Compared with the Period Average

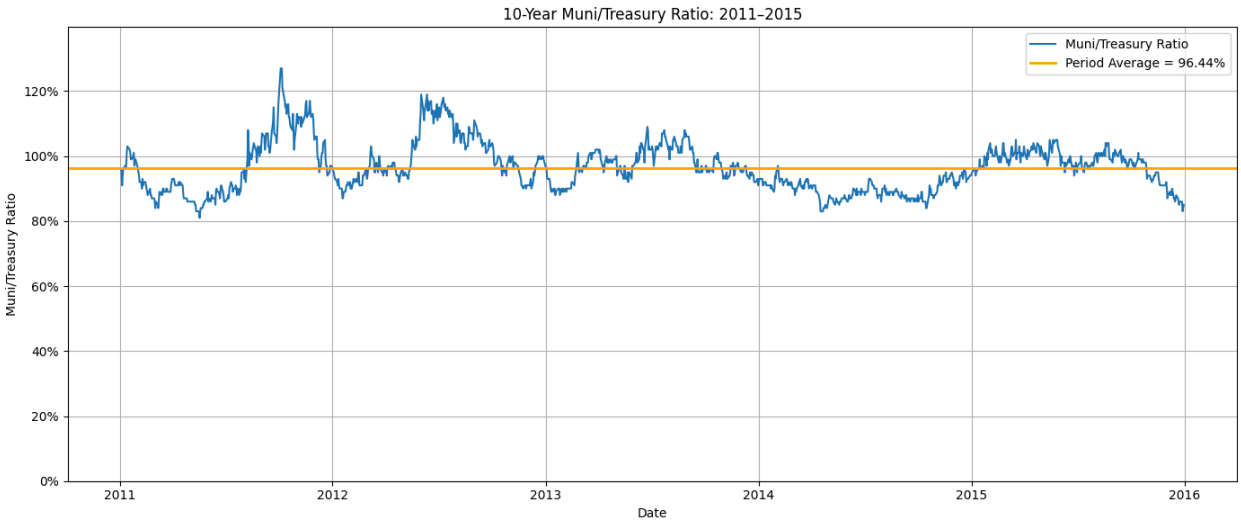


Figure 4: Municipal-to-Treasury Ratio, 2011-2015, Compared with the Period Average

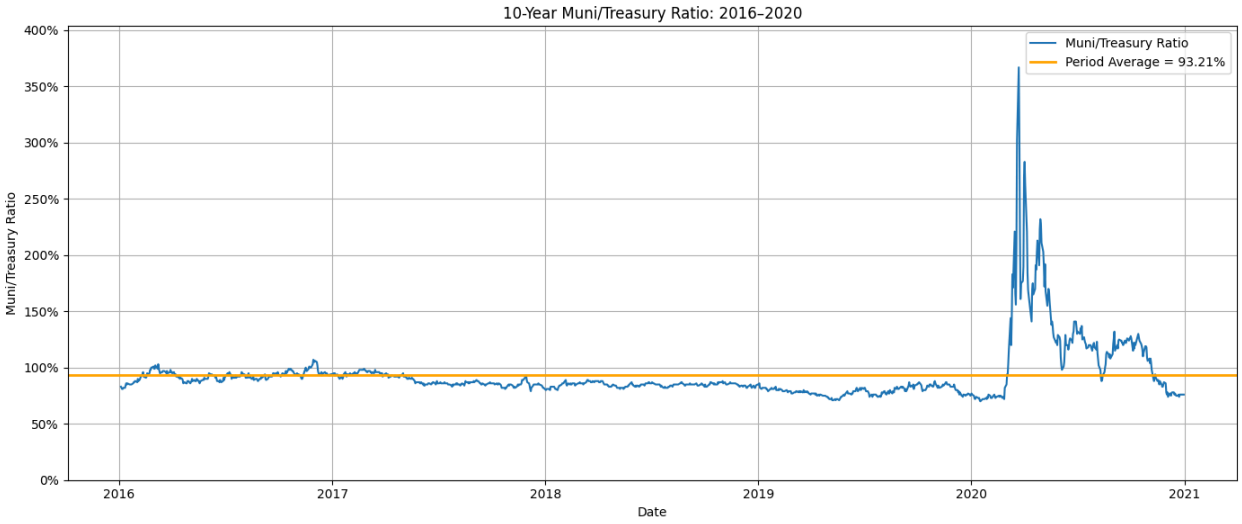


Figure 5: Municipal-to-Treasury Ratio, 2016-2020, Compared with the Period Average

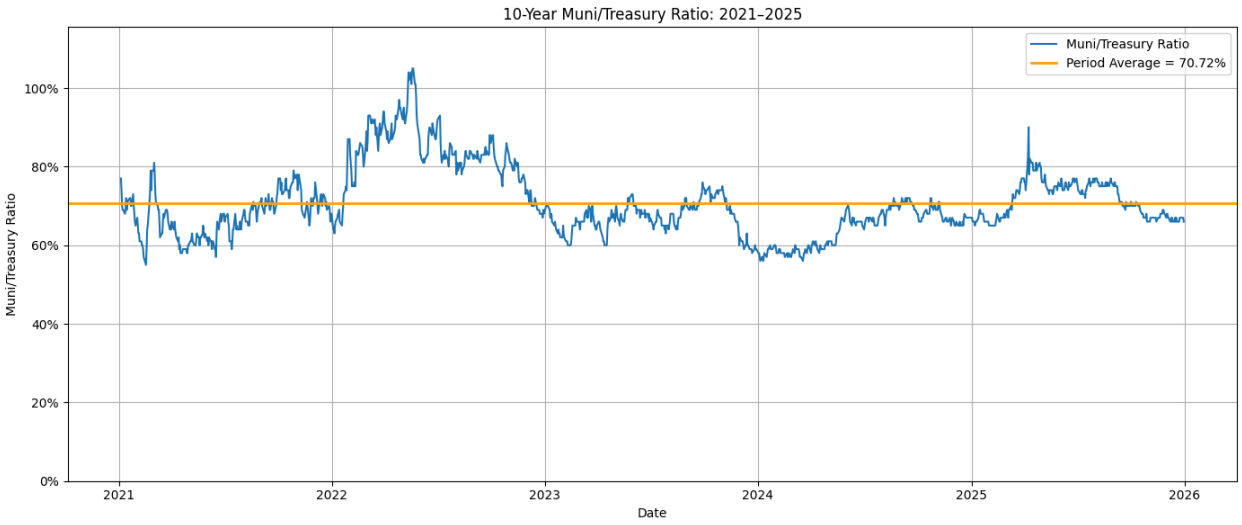


Figure 6: Municipal-to-Treasury Ratio, 2021-2025, Compared with the Period Average

In Figure 7, I examine what happened to municipal and Treasury bond returns after selected dates during these market dislocation periods. For each five-year period, my code

randomly selected one date from the identified dislocation window. This was done so that a date was not chosen simply because it produced the best return.

Regime	Start	Period	End	MMD Yield	Muni Return	Muni $ Return	Treasury Yield	Treasury Return	Treasury $ Return	Excess Return	Excess $ Return
2006–2010	1/13/09	1M	2/13/09	2.98% → 2.88%	1.1%	$11,042	2.33% → 2.89%	-4.6%	–$46,056	5.7%	$57,097
		2M	3/13/09	2.98% → 3.33%	-2.4%	–$24,206	2.33% → 2.89%	-4.4%	–$43,772	2.0%	$19,566
		3M	4/13/09	2.98% → 3.13%	-0.5%	–$5,099	2.33% → 2.88%	-4.1%	–$40,661	3.6%	$35,561
2011–2015	10/17/11	1M	11/17/11	2.53% → 2.29%	2.3%	$23,282	2.18% → 1.96%	2.2%	$21,549	0.2%	$1,732
		2M	12/16/11	2.53% → 1.93%	5.8%	$57,708	2.18% → 1.86%	3.2%	$32,258	2.5%	$25,451
		3M	1/17/12	2.53% → 1.69%	8.2%	$81,528	2.18% → 1.87%	3.3%	$32,950	4.9%	$48,578
2016–2020	5/15/20	1M	6/15/20	1.01% → 0.85%	1.6%	$16,026	0.64% → 0.71%	-0.6%	–$6,159	2.2%	$22,185
		2M	7/15/20	1.01% → 0.78%	2.3%	$23,411	0.64% → 0.64%	0.1%	$1,066	2.2%	$22,346
		3M	8/14/20	1.01% → 0.66%	3.6%	$35,520	0.64% → 0.71%	-0.5%	–$4,984	4.1%	$40,505
2021–2025	11/29/22	1M	12/29/22	2.74% → 2.63%	1.2%	$11,818	3.75% → 3.83%	-0.3%	–$3,449	1.5%	$15,267
		2M	1/30/23	2.74% → 2.19%	5.3%	$52,969	3.75% → 3.55%	2.3%	$22,692	3.0%	$30,277
		3M	2/28/23	2.74% → 2.59%	2.0%	$19,679	3.75% → 3.92%	-0.4%	–$4,336	2.4%	$24,016

Figure 7: Municipal and Treasury Forward Returns Following Selected Market Dislocations for a Hypothetical $1 Million Investment

The selected date was then used to calculate municipal and Treasury returns over future holding periods. For the municipal side, the returns represented a hypothetical AAA municipal bond priced using the MMD benchmark yield. A hypothetical $1 million investment was used for both the municipal and Treasury bonds so that the results could be compared in dollar and percentage terms.

The results varied across the four periods. In the 2006-2010 example, the municipal bond produced a 1.1% return after one month compared with a -4.6% Treasury return, giving the municipal bond a 5.7% advantage. However, the municipal return later became negative, showing that outperformance did not continue across every holding period.

In the 2011-2015 example, municipal returns increased from 2.3% after one month to 8.2% after three months, while Treasury returns ranged from 3.4% to 7.6%. The results changed

depending on the holding period rather than showing the same level of outperformance at every point.

During the 2016-2020 example, the municipal bond outperformed the Treasury bond across the three holding periods shown in Figure 7. At the three-month point, the municipal return was 4.1% higher than the Treasury return.

The 2021-2025 example also showed municipal returns above Treasury returns during the periods tested. However, the municipal return declined from 5.3% after two months to 2.0% after three months. This showed that even when municipal bonds performed better than Treasury bonds, the size of the return could change significantly over time.

Because Figure 7 uses only one randomly selected date from each dislocation period, these results should be viewed as examples rather than proof that every date would produce the same result. Running the analysis again could select different dates and produce different return values. A larger test using more dates would provide stronger evidence of how often municipal bonds outperform Treasury bonds after the muni-ratio becomes unusually high.

The final part of the results examined mean reversion, or how long it took for the Muni-to-Treasury ratio to move back toward its five-year average after a market dislocation. Figure 8 shows that the amount of time for the muni-ratio to revert to its period mean varied across the four periods.

Period	Period Mean	Number of Mean Reversion Events	Average Days to Mean	Median Days to Mean	Average Months to Mean	Fastest Return (Days)	Longest Return (Days)
2006–2010	89.75%	19	30.0	4.0	1.0	1	447
2011–2015	96.44%	44	19.4	5.5	0.6	1	156
2016–2020	93.19%	25	20.0	5.0	0.7	1	161
2021–2025	70.73%	29	23.9	3.0	0.8	1	322

Figure 8: Average Time for the Muni-to-Treasury Ratio to Return Toward Its Five-Year Mean

The average time to return toward the mean was approximately 30.0 days during 2006-2010, 19.4 days during 2011-2015, 20.0 days during 2016-2020, and 23.9 days during 2021-2025. These results showed that although the ratio often moved back toward its average after a large increase, there was no consistent amount of time for this process to occur.

5. Discussion

5.1 Interpretation of Results

The results show that the Muni-to-Treasury ratio can help identify periods when municipal bonds may offer better relative value compared with Treasury bonds. When the ratio moved significantly above its five-year average, municipal yields were higher compared to Treasury yields. In several of the examples tested in Figure 7, municipal bonds later produced higher returns than Treasury bonds. However, the results also showed that a high muni-ratio did not guarantee a positive return or consistent outperformance.

An important finding was that the normal level of the muni-ratio changed over time. The long-term historical average of approximately 87% did not represent all market environments equally well. The averages from 2006 through 2020 were generally above 90%, while the average from 2021 through 2025 was closer to 70%. This suggests that the level considered "high" or "low" depends partly on the economic and interest-rate environment. Using separate five-year averages therefore provided a better comparison than using one average for the entire dataset.

The different market conditions during each five-year period also help explain why the ratio behaved differently. During the 2006-2010 period, the Great Financial Crisis caused investors to move toward Treasury securities, which pushed Treasury yields lower. At the same

time, concerns about state and local government finances kept municipal yields higher, causing the muni-ratio to rise sharply.

The 2011-2015 period occurred during the economic recovery following the GFC. Although market conditions improved, concerns about municipal credit quality remained. For example, predictions of widespread municipal defaults created additional concern among investors. These fears did not lead to the large number of defaults that had been predicted, but they contributed to continued attention on municipal credit risk.

During the 2016-2020 period, the COVID-19 pandemic created another major market disruption. Treasury yields declined sharply as investors moved toward safer investments and the Federal Reserve lowered interest rates. Municipal bonds were also affected by concerns about falling state and local government revenues. These conditions contributed to the large increase in the muni-ratio during this period.

The 2021-2025 period looked different from the earlier periods. As the economy recovered from the pandemic and inflation increased, the Federal Reserve raised interest rates. Both municipal and Treasury yields adjusted to the new interest-rate environment, and the average muni-ratio moved closer to 70%. This helps show why the same historical average may not be appropriate for every market period.

A key finding was that the muni-ratio should not be used by itself to predict bond returns. The ratio compares municipal yields with Treasury yields, which means that it can remain relatively stable even when both yields are changing. For example, if municipal and Treasury yields both decline by similar amounts, the ratio may show very little change even though bond prices are increasing. An investor looking only at the ratio could therefore miss important changes in the underlying bonds.

The forward return analysis shows the difference between identifying relative value and predicting future returns. A high muni-ratio may suggest that municipal bonds are relatively attractive compared with Treasury bonds, but it does not mean that municipal bond prices will immediately increase. Returns depend on what happens to yields after the investment is made. Since bond prices and yields move in opposite directions, rising yields can result in negative returns even if the muni-ratio initially appeared attractive.

The ratio generally moved back toward its five-year average after large dislocations, but the amount of time needed varied across the four periods. This means that the muni-ratio may identify an opportunity without showing exactly when that opportunity will produce a return. An investor may have to wait several days or months for the relationship between municipal and Treasury yields to move back toward its normal level.

Taxes are an important factor when comparing municipal and Treasury bond returns. Municipal bond interest is generally exempt from federal income taxes, while Treasury interest is subject to federal income tax. In this analysis, I assumed a 30% federal income tax rate to make an after-tax comparison. This tax advantage can make municipal bonds attractive even when their stated yield is below the Treasury yield. However, the actual tax benefit depends on the investor's individual tax rate, so the 30% assumption should be viewed as a simplified example. Overall, the results generally support my hypothesis that a muni-ratio significantly above its five-year average can help identify periods when municipal bonds offer better relative value compared to Treasury bonds. However, the results do not show that a high ratio will always lead to positive returns or that municipal bonds will always outperform Treasury bonds. The muni-ratio appears to be more useful as a relative value indicator than as a precise predictor of

future returns or market timing. Therefore, it should be used together with other information, including interest rates, taxes, credit conditions, market conditions, and investor demand.

5.2 Limitations of the Analysis

There were several limitations in this study that are important to consider when looking at the results. First, the analysis only used the 10-year Triple-A MMD curve[13] and the 10-year U.S. Treasury yield[14,15,16,17]. This made it possible to compare the same maturity across the two markets, but it also means that the results may not apply in the same way to shorter- or longer-term bonds. Various parts of the yield curve can behave differently depending on interest rates, market conditions, and investor demand. Future research could look at other maturities, such as the 3-year, 5-year, 7-year, 20-year, or 30-year curves, to see if the same patterns continue. Another limitation was the use of a 30% federal income tax rate when comparing municipal bonds with Treasury bonds. This was used as a simple assumption to simplify the comparison, but every investor has a different tax situation. Some investors may have higher tax rates, while others may have lower ones. Because the tax advantage is one of the main reasons investors buy municipal bonds, changing the tax rate could affect how attractive municipal bonds appear compared to Treasury bonds.

The way the market dislocation periods were selected was also a limitation. I focused on certain time periods when the muni-ratio moved well above its five-year average and then looked at selected observations within those periods. This helped show how the ratio behaved after major changes in the market, but the results may depend on which dates were chosen. A larger

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[13] LSEG, “The Municipal Market Monitor (TM3),”
[14] U.S. Department of the Treasury, 1-2-2024, “Daily Treasury Rates,”
[15] U.S. Department of the Treasury, 1-2-2025, “Daily Treasury Rates,”
[16] U.S. Department of the Treasury, 1-2-2026, “Daily Treasury Rates,”
[17] U.S. Department of the Treasury, 12-29-2023, “Daily Treasury Rate Archives,”

study using every date that met a specific condition, rather than selected dates, would provide a stronger test of whether a high muni-ratio consistently leads to better relative returns.

The analysis also showed that the muni-ratio does not explain everything that affects bond performance. Municipal and Treasury yields can move because of a variety of factors, including interest rates, inflation, credit risk, tax policy, Federal Reserve decisions, investor demand, and major economic events. The ratio only shows the relationship between the two yields, so it may stay relatively unchanged even when both yields are moving significantly. Because of this, the ratio should be viewed as one piece of information rather than a complete prediction of future returns.

Finally, this study focused mainly on historical relationships. While past market data can help identify patterns, the same pattern may not always repeat in the future. Every financial crisis and economic environment is different, which can affect how quickly the muni-ratio moves back toward its average and whether municipal bonds outperform Treasury bonds. For this reason, the results of this study should be viewed as evidence that the ratio can help identify possible relative value opportunities, rather than proof that it will work the same way in every future market.

5.3 Comparison to Existing Literature

The muni-ratio is useful because municipal and Treasury bonds are affected by some of the same economic conditions, but they are also impacted by different factors. Previous research shows that taxes, credit risk, liquidity, investor demand, and major financial events can all influence municipal bond yields. Understanding these factors is important when studying whether changes in the muni-ratio can help identify relative value opportunities.

One important factor in municipal bond pricing is the tax advantage that municipal bonds provide. Interest earned on most municipal bonds is exempt from federal income taxes, while

interest from Treasury bonds is subject to federal income taxes. Babina et al. (2021)[18] studied how differences in tax policies affect municipal bond ownership, demand, and yields. Their research found that tax advantages can influence which municipal bonds investors choose to own and how sensitive their yields are to changes in supply and risk. This is important for my research because the tax advantage of municipal bonds is one reason an investor may accept a lower municipal yield compared to a Treasury yield. Therefore, simply comparing the two yields without considering taxes may not demonstrate the full relative value of the two investments.

Another important theme in previous research is that the municipal bond market can be affected by liquidity, credit risk, and changes in investor demand. Cestau et al. (2019)[19] explain that municipal bond prices and yields are influenced by several factors, including taxes, credit quality, liquidity, and the structure of the municipal market. They also discuss how municipal bonds trade in a decentralized market, where prices can sometimes vary between transactions. These factors can cause municipal yields to change differently from Treasury yields. This is important for the muni-ratio because a change in the ratio may not always be caused by a change in interest rates. It could also be impacted by investors becoming more concerned about municipal credit risk, a decrease in demand for municipal bonds, or other problems within the municipal market.

Major financial crises can make these differences even more noticeable. Li and Lu (2020)[20] studied the municipal bond market during the COVID-19 pandemic and found that municipal offering yields increased as COVID-19 cases and deaths increased, while the number of new municipal bond issues decreased. They also found that local emergency declarations were

[18] Babina, T., Jotikasthira, C., Lundblad, C., & Ramadorai, T. (2021). Heterogeneous taxes and limited risk sharing: Evidence from municipal bonds.
[19] Cestau, D., Hollifield, B., Li, D., & Schürhoff, N. (2019). Municipal bond markets.
[20] Li, Tao, and Lu, Jing. Municipal Finance During the COVID-19 Pandemic

associated with an increase of about 69 basis points in municipal offering yields. Investors became more careful about certain types of municipal bonds, including transportation bonds and bonds issued by financially weaker states. The Federal Reserve later stepped in to support the municipal market, which helped calm some of the market disruption. This research shows how a major economic event can affect municipal bonds differently from Treasury bonds and cause the relationship between their yields to change.

These previous studies help explain some of the reasons the muni-ratio can move over time. Taxes can affect investor demand, concerns about credit and liquidity can push municipal yields higher, and major financial events can cause large differences between municipal and Treasury yields. However, the studies mainly focus on what affects municipal bond prices and yields rather than testing whether the muni-ratio itself can be used to identify future relative value opportunities.

My research builds on these ideas by focusing specifically on the relationship between AAA municipal yields and Treasury yields. I examine whether a muni-ratio that moves above its five-year average can help identify periods when municipal bonds offer better relative value and whether municipal bonds later outperform Treasury bonds. I also examine how long it takes the ratio to move back toward its average after a major increase. This allows the study to look not only at why the ratio changes, but also at whether those changes can be useful when identifying possible investment opportunities.

6. Conclusion

The goal of this study was to find out whether the muni-ratio can help identify times when AAA municipal bonds may offer better value than U.S. Treasury bonds. My analysis

shows that the ratio can be an useful indicator, especially when it rises well above its five-year average. I also found that using one long-term historical average was not always the best way to judge whether the ratio was high or low. The normal level of the ratio changed over time, so using separate five-year averages gave a better picture of what was happening in each market environment. The ratio rose above its five-year average during major market events such as the Great Financial Crisis and the COVID-19 pandemic, when the relationship between municipal and Treasury yields changed significantly.

At the same time, the results showed that the muni-ratio is not a perfect way to predict future returns. A high ratio may suggest that municipal bonds are relatively attractive, but it does not mean that they will always produce higher returns. Municipal and Treasury yields can both move in the same direction, which can cause the ratio to stay almost the same even while bond prices are changing. This means that the ratio can give investors useful information, but it cannot tell the full story on its own.

Another important result was that the ratio did not return to its average at the same speed in every period. In some periods, mean reversion happened fairly quickly, while in others it took much longer. This showed me that finding a possible investment opportunity is not the same as knowing when that opportunity will actually pay off. Market conditions, interest rates, investor demand, and other economic factors can all affect how long the process takes.

My research supports the idea that the muni-ratio can be a helpful tool for identifying relative value opportunities between municipal bonds and Treasury bonds. However, it should not be used as the only factor in making an investment decision. The ratio is most useful when it is considered along with changes in interest rates, taxes, market conditions, credit risk, and investor demand. Future research could build on this study by looking at more bond maturities,

testing more market periods, and using a larger number of observations to determine whether these results remain consistent.

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